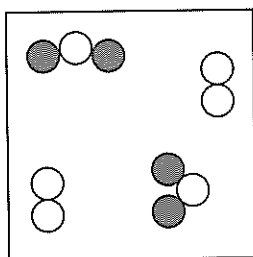


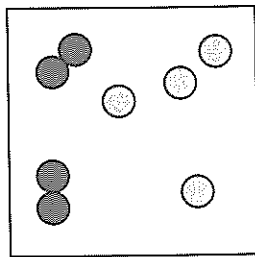
Name: _____

Skills: interpreting, understanding

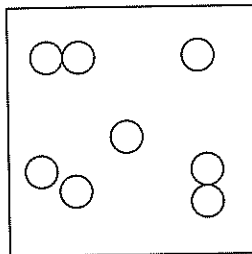
Examine the four boxes below. Each circle represents an atom and the shading pattern represents different types of atoms.



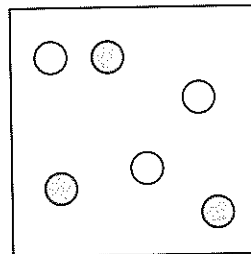
A



B



C



5

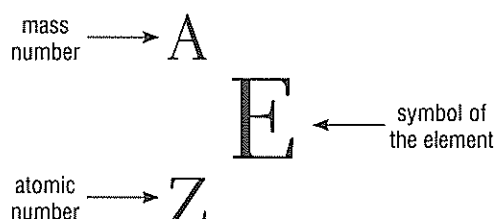
Questions

- Which of the boxes contains the following (note: it is possible that none of the boxes contain the requested item—if this is the case then write 'none'. It is also possible that more than one box may contain the requested item):
 - individual atoms of elements only? _____
 - chemical compounds only? _____
 - chemical elements only? _____
 - a mixture of compounds and elements? _____
- When two atoms join together we can use the term 'diatomic molecule' to describe the molecule formed (from di = 2). When a chemical element exists as single atoms, the term 'monatomic' (from mon = 1) is used. Using the samples in boxes A to D above, draw examples of the following:
 - two different types of diatomic elemental molecules
 - two different monatomic elements
 - a chemical compound

Name: _____

There has to be an EAZ ier way to represent atoms!

There is an easy way to represent atoms so that we can describe the number of protons, neutrons and electrons in any nucleus. This method is summarised by using the letters E to represent the element symbol, A to represent the mass number (number of protons and neutrons in the nucleus) and the atomic number (number of protons).



Use this method to complete the last column in the following table. A couple of examples have been done for you.

Atom	Atomic number (number of protons)	Number of neutrons	Mass number	${}^A_Z\text{E}$
Hydrogen	1	0	1	
Hydrogen-2 (deuterium)	1	1	2	${}^2_1\text{H}$
Carbon	6	6	12	
Oxygen	8	8	16	
Lead	82	126	208	${}^{208}_{82}\text{Pb}$
Uranium	92	146	238	

Extension

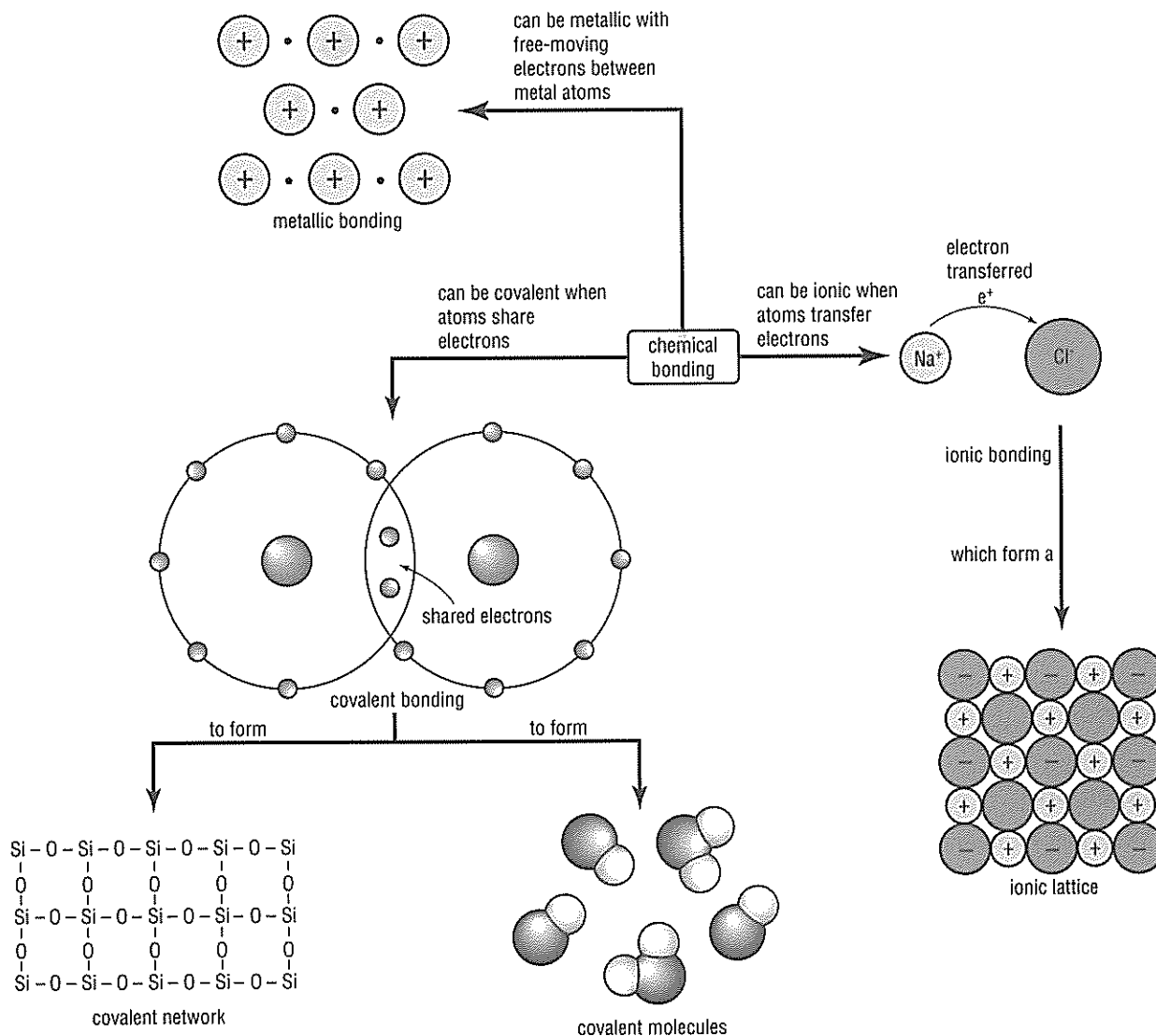
Given that for neutral atoms the number of electrons in the electron cloud is equal to the number of protons in the nucleus, add an extra column to the end of the table above to include the number of electrons you would find in a neutral atom in the examples above.



Skills: interpreting, knowledge

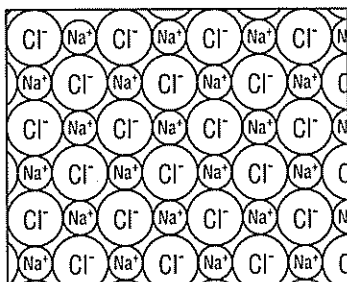
The concept map below summarises the bonding types explored in Focus 5.7. Review it carefully before you start. You may like to refer to Focus 5.7 when looking through the concept map.

On the next page are some diagrams representing a range of substances. For each of the diagrams you are to identify the type of bonding as ionic, metallic, covalent network or covalent molecules. There is also some space under your answer to write a supporting statement for your decision. Discuss your answers with your teacher and class.

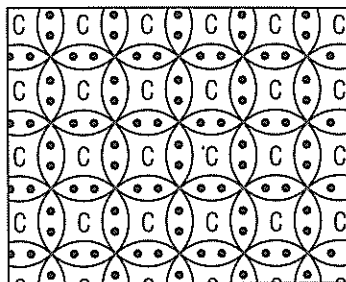


Name: _____

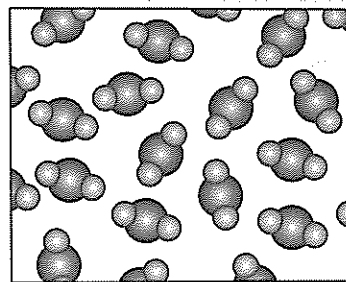
sodium chloride lattice



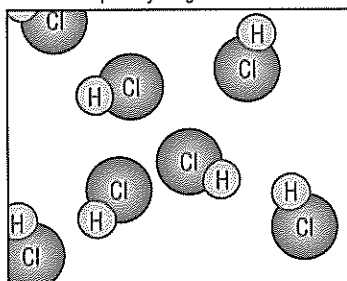
diamond lattice



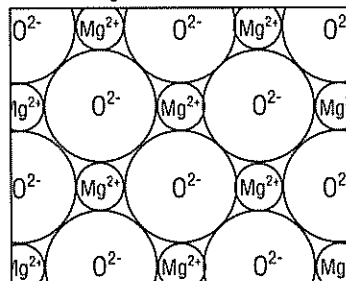
liquid water



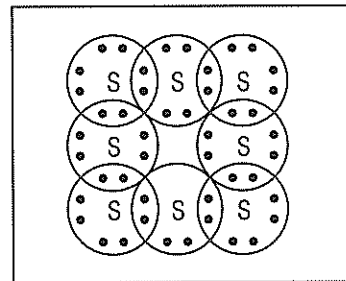
liquid hydrogen chloride



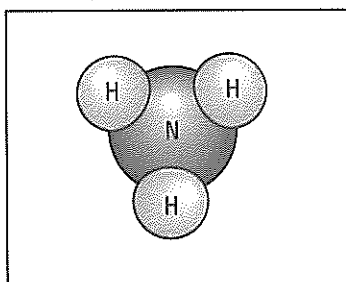
magnesium oxide lattice



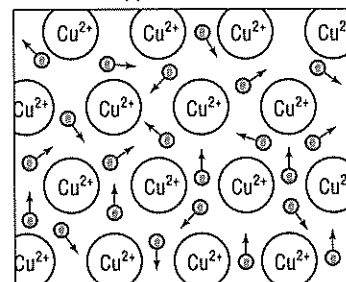
sulfur molecule



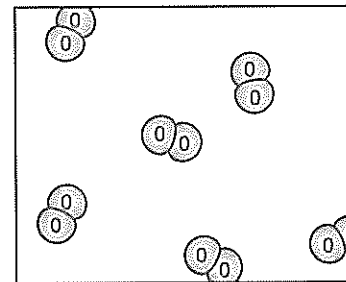
ammonia molecule



copper metal lattice



oxygen gas





Skill: knowledge

Atomic number	Symbol	Name	Atomic weight	Atomic radius*
1	H	hydrogen	1	37
2	He	helium	4	90
3	Li	lithium	7	125
4	Be	beryllium	9	90
5	B	boron	11	82
6	C	carbon	12	77
7	N	nitrogen	14	75
8	O	oxygen	16	73
9	F	fluorine	19	72
10	Ne	neon	20	71
11	Na	sodium	23	154
12	Mg	magnesium	24	136
13	Al	aluminum	27	118
14	Si	silicon	28	110
15	P	phosphorus	31	106
16	S	sulfur	32	102
17	Cl	chlorine	35	99
18	Ar	argon	40	95
19	K	potassium	39	203
20	Ca	calcium	40	174
21	Sc	scandium	45	144
22	Ti	titanium	48	132
23	V	vanadium	51	122
24	Cr	chromium	52	118
25	Mn	manganese	55	117
26	Fe	iron	56	117
27	Co	cobalt	59	116
28	Ni	nickel	59	115
29	Cu	copper	64	117
30	Zn	zinc	65	125
31	Ga	Gallium	70	126
32	Ge	germanium	73	122
33	As	arsenic	75	120
34	Se	selenium	79	117
35	Br	bromine	80	114
36	Kr	krypton	84	112
37	Rb	rubidium	85	216
38	Sr	strontium	88	191
39	Y	yttrium	89	162
40	Zr	zirconium	91	145
41	Nb	niobium	93	134
42	Mo	molybdenum	96	130
43	Tc	technetium	98	127
44	Ru	ruthenium	101	134
45	Rh	rhodium	103	125
46	Pd	palladium	106	128
47	Ag	silver	108	134
48	Cd	cadmium	112	148
49	In	indium	115	144
50	Sn	tin	119	141
51	Sb	antimony	122	140
52	Te	tellurium	128	136
53	I	iodine	127	133
54	Xe	xenon	131	131
55	Cs	cesium	133	235
56	Ba	barium	137	198
57	La	lanthanum	139	169
58	Ce	cerium	140	165
59	Pr	praseodymium	141	165

Atomic number	Symbol	Name	Atomic weight	Atomic radius*
60	Nd	neodymium	144	164
61	Pm	promethium	145	163
62	Sm	samarium	150	162
63	Eu	europium	152	185
64	Gd	gadolinium	157	161
65	Tb	terbium	159	159
66	Dy	dysprosium	163	159
67	Ho	holmium	165	158
68	Er	erbium	167	157
69	Tm	thulium	169	156
70	Yb	ytterbium	173	144
71	Lu	lutetium	175	134
72	Hf	hafnium	178	136
73	Ta	tantalum	181	128
74	W	tungsten	184	126
75	Re	rhenium	186	127
76	Os	osmium	190	128
77	Ir	iridium	192	126
78	Pt	platinum	195	129
79	Au	gold	197	134
80	Hg	mercury	201	149
81	Tl	thallium	204	148
82	Pb	lead	207	146
83	Bi	bismuth	209	145
84	Po	polonium	209	
85	At	astatine	210	
86	Rn	radon	222	
87	Fr	francium	223	
88	Ra	radium	226	
89	Ac	actinium	227	
90	Th	thorium	232	
91	Pa	protactinium	231	
92	U	uranium	238	
93	Np	neptunium	237	
94	Pu	plutonium	244	
95	Am	americium	243	
96	Cm	curium	247	
97	Bk	berkelium	247	
98	Cf	californium	251	
99	Es	einsteinium	252	
100	Fm	fermium	257	
101	Md	mendelevium	258	
102	No	nobelium	259	
103	Lr	lawrencium	262	
104	Rf	rutherfordium	261	
105	Db	dubnium	262	
106	Sg	seaborgium	266	
107	Bh	bohrium	264	
108	Hs	hassium	277	
109	Mt	meitnerium	268	
110	Ds	darmstadtium	271	
111	Rg	roentgenium	272	
112	Uub	ununbium	285	

*Atomic radii are in picometres (pm).

The elements—page 2

Name: _____

Questions

Use the data on the previous page to answer these questions.

1 Give the symbols for the following elements:

- tungsten _____
- sodium _____
- zinc _____
- einsteinium _____
- tin _____
- boron _____
- cerium _____
- radon _____

2 Give the names for the following elements:

- Am _____
- Uub _____
- Ne _____
- Ar _____
- Cd _____
- O _____
- H _____
- Si _____
- S _____

3 Identify the following atoms from the table, giving their names and symbols:

- a the atom with an atomic number of 88

- b the atom with an atomic weight of 39

- c the atom with an atomic radius of 133

- d the atom with the symbol Hg

- e the atom with an atomic number of 16

- f the atom with the symbol He

- g the atom with atomic weight 108

4 The atomic weight of an atom is its mass compared with that of a carbon atom. If an atom has an atomic weight less than carbon it is lighter than carbon. If an atom has an atomic weight larger than carbon it is heavier than carbon.

a State the atomic weight of a carbon atom.

b Identify two atoms that are lighter than carbon.

c Identify two atoms that are heavier than carbon.

d Describe what happens to the atomic weight as the atomic number increases.

e Calculate how many times heavier a magnesium atom is than carbon.

f Calculate how many times lighter a helium atom is than carbon.

5 The atomic radius gives an indication of the size of an atom. The table does not give the atomic radius for all atoms. Use the data shown to give the name and symbol for the:

a smallest atom _____

b largest atom _____



Skills: interpreting, knowledge

Group I		Group II		Group III										Group IV		Group V		Group VI		Group VII		Group VIII																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																	
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H hydrogen 1		He helium 2		Li lithium 3		Be beryllium 4		B boron 5		C carbon 6		N nitrogen 7		O oxygen 8		F fluorine 9		Ne neon 10		Na sodium 11		Mg magnesium 12		Al aluminum 13		Si silicon 14		P phosphorus 15		S sulfur 16		Cl chlorine 17		Ar argon 18		K potassium 19		Ca calcium 20		Sc scandium 21		Ti titanium 22		V vanadium 23		Cr chromium 24		Mn manganese 25		Fe iron 26		Co cobalt 27		Ni nickel 28		Cu copper 29		Zn zinc 30		Ga gallium 31		Ge germanium 32		As arsenic 33		Se selenium 34		Br bromine 35		Kr krypton 36		Rb rubidium 37		Sr strontium 38		Y yttrium 39		Zr zirconium 40		Nb niobium 41		Mo molybdenum 42		Tc technetium 43		Ru ruthenium 44		Rh rhodium 45		Pd palladium 46		Ag silver 47		Cd cadmium 48		In indium 49		Sn tin 50		Sb antimony 51		Te tellurium 52		I iodine 53		Xe xenon 54		Cs cesium 55		Ba barium 56		La ^a lanthanum 57		Hf hafnium 72		Ta tantalum 73		W tungsten 74		Re rhenium 75		Os osmium 76		Ir iridium 77		Pt platinum 78		Au gold 79		Hg mercury 80		Tl thallium 81		Pb lead 82		Bi bismuth 83		Po polonium 84		At astatine 85		Rn radon 86		Fr francium 87		Ra radium 88		Ac ^{a*} actinium 89		Rf rutherfordium 104		Db dubnium 105		Sg seaborgium 106		Bh bohrium 107		Hs hassium 108		Mt meitnerium 109		Ds darmstadtium 110		Rg roentgenium 111																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																			
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98		Period 99		Period 100		Period 101		Period 102		Period 103		Period 104		Period 105		Period 106		Period 107		Period 108		Period 109		Period 110		Period 111		Period 112		Period 113		Period 114		Period 115		Period 116		Period 117		Period 118		Period 119		Period 120		Period 121		Period 122		Period 123		Period 124		Period 125		Period 126		Period 127		Period 128		Period 129		Period 130		Period 131		Period 132		Period 133		Period 134		Period 135		Period 136		Period 137		Period 138		Period 139		Period 140		Period 141		Period 142		Period 143		Period 144		Period 145		Period 146		Period 147		Period 148		Period 149		Period 150		Period 151		Period 152		Period 153		Period 154		Period 155		Period 156		Period 157		Period 158		Period 159		Period 160		Period 161		Period 162		Period 163		Period 164		Period 165		Period 166		Period 167		Period 168		Period 169		Period 170		Period 171		Period 172		Period 173		Period 174		Period 175		Period 176		Period 177		Period 178		Period 179		Period 180		Period 181		Period 182		Period 183		Period 184		Period 185		Period 186		Period 187		Period 188		Period 189		Period 190		Period 191		Period 192		Period 193		Period 194		Period 195		Period 196		Period 197		Period 198		Period 199		Period 200		Period 201		Period 202		Period 203		Period 204		Period 205		Period 206		Period 207		Period 208		Period 209		Period 210		Period 211		Period 212		Period 213		Period 214		Period 215		Period 216		Period 217		Period 218		Period 219		Period 220		Period 221		Period 222		Period 223		Period 224		Period 225		Period 226		Period 227		Period 228		Period 229		Period 230		Period 231		Period 232		Period 233		Period 234		Period 235		Period 236		Period 237		Period 238		Period 239		Period 240		Period 241		Period 242		Period 243		Period 244		Period 245		Period 246		Period 247		Period 248		Period 249		Period 250		Period 251		Period 252		Period 253		Period 254		Period 255		Period 256		Period 257		Period 258		Period 259		Period 260		Period 261		Period 262		Period 263		Period 264		Period 265		Period 266		Period 267		Period 268		Period 269		Period 270		Period 271		Period 272		Period 273		Period 274		Period 275		Period 276		Period 277		Period 278		Period 279		Period 280		Period 281		Period 282		Period 283		Period 284		Period 285		Period 286		Period 287		Period 288		Period 289		Period 290		Period 291		Period 292		Period 293		Period 294		Period 295		Period 296		Period 297		Period 298		Period 299		Period 300		Period 301		Period 302		Period 303		Period 304		Period 305		Period 306		Period 307		Period 308		Period 309		Period 310		Period 311		Period 312		Period 313		Period 314		Period 315		Period 316		Period 317		Period 318		Period 319		Period 320		Period 321		Period 322		Period 323		Period 324		Period 325		Period 326		Period 327		Period 328		Period 329		Period 330		Period 331		Period 332		Period 333		Period 334		Period 335		Period 336		Period 337		Period 338		Period 339		Period 340		Period 341		Period 342		Period 343		Period 344		Period 345		Period 346		Period 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430		Period 431		Period 432		Period 433		Period 434		Period 435		Period 436		Period 437		Period 438		Period 439		Period 440		Period 441		Period 442		Period 443		Period 444		Period 445		Period 446		Period 447		Period 448		Period 449		Period 450		Period 451		Period 452		Period 453		Period 454		Period 455		Period 456		Period 457		Period 458		Period 459		Period 460		Period 461		Period 462		Period 463		Period 464		Period 465		Period 466		Period 467		Period 468		Period 469		Period 470		Period 471		Period 472		Period 473		Period 474		Period 475		Period 476		Period 477		Period 478		Period 479		Period 480		Period 481		Period 482		Period 483		Period 484		Period 485		Period 486		Period 487		Period 488		Period 489		Period 490		Period 491		Period 492		Period 493		Period 494		Period 495		Period 496		Period 497		Period 498		Period 499		Period 500		Period 501		Period 502		Period 503		Period 504		Period 505		Period 506		Period 507		Period 508		Period 509		Period 510		Period 511		Period 512		Period 513		Period 514		Period 515		Period 516		Period 517		Period 518		Period 519		Period 520		Period 521		Period 522		Period 523		Period 524		Period 525		Period 526		Period 527		Period 528		Period 529		Period 530		Period 531		Period 532		Period 533		Period 534		Period 535		Period 536		Period 537		Period 538		Period 539		Period 540		Period 541		Period 542		Period 543		Period 544		Period 545		Period 546		Period 547		Period 548		Period 549		Period 550		Period 551		Period 552		Period 553		Period 554		Period 555		Period 556		Period 557		Period 558		Period 559		Period 560		Period 561		Period 562		Period 563		Period 564		Period 565		Period 566		Period 567		Period 568		Period 569		Period 570		Period 571		Period 572		Period 573		Period 574		Period 575		Period 576		Period 577		Period 578		Period 579		Period 580		Period 581		Period 582		Period 583		Period 584		Period 585		Period 586		Period 587		Period 588		Period 589		Period 590		Period 591		Period 592		Period 593		Period 594		Period 595		Period 596		Period 597		Period 598		Period 599		Period 600		Period 601		Period 602		Period 603		Period 604		Period 605		Period 606		Period 607		Period 608		Period 609		Period 610		Period 611		Period 612		Period 613		Period 614		Period 615		Period 616		Period 617		Period 618		Period 619		Period 620		Period 621		Period 622		Period 623		Period 624		Period 625		Period 626		Period 627		Period 628		Period 629		Period 630		Period 631		Period 632		Period 633		Period 634		Period 635		Period 636		Period 637		Period 638		Period 639		Period 640		Period 641		Period 642		Period 643		Period 644		Period 645		Period 646		Period 647		Period 648		Period 649		Period 650		Period 651		Period 652		Period 653		Period 654		Period 655		Period 656		Period 657		Period 658		Period 659		Period 660		Period 661		Period 662		Period 663		Period 664		Period 665		Period 666		Period 667		Period 668		Period 669		Period 670		Period 671		Period 672		Period 673		Period 674		Period 675		Period 676		Period 677		Period 678		Period 679		Period 680		Period 681		Period 682		Period 683		Period 684		Period 685		Period 686		Period 687		Period 688		Period 689		Period 690		Period 691		Period 692		Period 693		Period 694		Period 695		Period 696		Period 697		Period 698		Period 699		Period 700		Period 701		Period 702		Period 703		Period 704		Period 705		Period 706		Period 707		Period 708		Period 709		Period 710		Period 711		Period 712		Period 713		Period 714		Period 715		Period 716		Period 717		Period 718		Period 719		Period 720		Period 721		Period 722		Period 723		Period 724		Period 725		Period 726		Period 727		Period 728		Period 729		Period 730		Period 731		Period 732		Period 733		Period 734		Period 735		Period 736		Period 737		Period 738		Period 739		Period 740		Period 741		Period 742		Period 743		Period 744		Period 745		Period 746		Period 747		Period 748		Period 749		Period 750		Period 751		Period 752		Period 753		Period 754		Period 755		Period 756		Period 757		Period 758		Period 759		Period 760		Period 761		Period 762		Period 763	

H — chemical symbol
hydrogen — name of element
1 — atomic number

The periodic table—page 2

Name: _____

Questions

1 On the periodic table, colour each of the following in different colours:

- a metals
- b non-metals
- c semi-metals (metalloids).

2 Identify which group consists of all gases.

3 What other name is given to the group VII elements?

4 State the atomic number for these elements:

- a platinum _____
- b gold _____
- c oxygen _____
- d magnesium _____
- e californium _____
- f bromine _____
- g krypton _____

5 State the name for atoms with the following atomic numbers:

- a 6 _____
- b 3 _____
- c 40 _____
- d 13 _____
- e 92 _____
- f 56 _____
- g 1 _____

6 a Identify three elements that start with a C.

b State the symbols for the elements you have chosen.

c Explain why some elements have one letter for their symbol and others have two.

7 Outline a reason why elements are placed in the same group.

8 Explain what the atomic number tells us about an element.

9 Identify three elements that you think are named after famous people, and propose who that person was.

Ionic compounds, names and formulas—page 1

**Skills: understanding, applying**

Use the following table and examples to answer the questions.

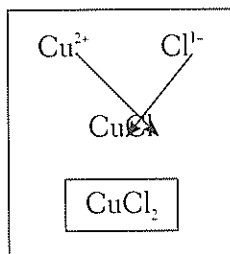
Positive ions (cations)		Negative ions (anions)	
Formula	Name	Formula	Name
Mg^{2+}	magnesium	Cl^-	chloride
Na^+	sodium	CO_3^{2-}	carbonate
NH_4^+	ammonium	OH^-	hydroxide
Al^{3+}	aluminium	O^{2-}	oxide
Ca^{2+}	calcium	N^{3-}	nitride
Li^+	lithium	NO_3^-	nitrate
Cu^{2+}	copper(II)	SO_4^{2-}	sulfate
Cr^{3+}	chromium(III)	Br^-	bromide

Remember, the compound has to have an overall charge of zero, or it will not be stable.

Examples

Write a balanced compound formula for:

- 1 copper (II) chloride



Start by writing the formulas.

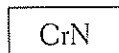
Put the ion formulas together, without the charges. Always put the positive ion first.

Cross-multiply, putting the number of the charge on one ion, near the other ion, as shown.

Balanced compound formula

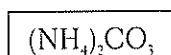
- 2 chromium (III) nitride

Following the above method, you end up with the formula Cr_3N_3 . This is an incorrect formula. If the numbers next to each ion are the same, just use one of each.



- 3 ammonium carbonate

If more than one of a polyatomic ion is needed to balance the formula, use brackets.



Ionic compounds, names and formulas—page 2

Name: _____

Questions

1 Name the following compounds:

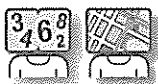
- a MgCl_2 _____
- b $(\text{NH}_4)_2\text{O}$ _____
- c Ca_3N_2 _____
- d CuCO_3 _____
- e $\text{Cr}_2(\text{SO}_4)_3$ _____
- f $\text{Cr}(\text{NO}_3)_3$ _____
- g $\text{Al}(\text{OH})_3$ _____

2 Write formulas for these compounds:

- a lithium carbonate _____
- b aluminium bromide _____
- c copper (II) nitrate _____
- d sodium chloride _____
- e calcium bromide _____
- f ammonium hydroxide _____
- g lithium nitride _____

3 Sodium chromate has the formula Na_2CrO_4 .

- a What is the formula of the chromate ion? _____
- b What is the formula of:
 - i magnesium chromate? _____
 - ii aluminium chromate? _____



Name: _____

Skills: knowledge, applying

Elements combine to form compounds. Each compound is represented by a unique chemical formula. A formula shows each element present, with subscripts stating how many atoms of each element are in the compound.

The formulas of molecules

- 1 Complete the following table with the names and formulas of common molecules that you should know. The atoms in these molecules are held together by covalent bonds.

Name	Formula	Name	Formula
nitrogen gas	N ₂	hydrogen gas	
oxygen gas		water	
carbon dioxide		sulfur dioxide gas	
	CO		SO ₃
	NO ₂	chlorine gas	
nitrogen monoxide			F ₂

The charges of ions

- 2 In order to write correct formulas for ionic compounds you need to know the charge on each ion involved. To help you learn these, complete the following tables of common ions.

Cations

+1		+2		+3	
	Na ⁺		Zn ²⁺	iron(III)	
hydrogen		copper		aluminium	
potassium		magnesium			
lithium			Ca ²⁺		
	Ag ⁺	iron(II)			

Anions

-1		-2		-3	
	Cl ⁻	oxide		nitride	
fluoride		sulfide		phosphate	
iodide			CO ₃ ²⁻		
bromide		sulfate			
nitrate					
	OH ⁻				
hydrogen carbonate (bicarbonate)					

Writing formulas—page 2

Name: _____

Ionic formulas and names

Practise writing the formulas and names for ionic compounds by completing the tables below. If you need to revise how to do this, see Focus 5.9, or read the hint below.

Hint: When ions come together to form compounds, they combine in a ratio that gives the compound a total charge of zero. There must be enough negative charges to balance the positive charges and vice versa.

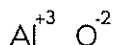
Sodium and chloride ions combine in a 1:1 ratio because sodium ions have a +1 charge and chloride ions have a -1 charge. Add these charges together: $+1 + (-1) = 0$. Thus, one of each ion join to give a compound with a total charge of zero. The formula is NaCl, and the name of this compound is sodium chloride.

Note that the charges of the ions are not included in compound formulas, but the numbers of each ion (the subscript numbers) are included. These subscript numbers indicate how many of each ion are in the formula. For instance, MgCl_2 indicates that there are one magnesium ion and two chloride ions in the formula. This is because magnesium (Mg) has a charge of +2 and chloride (Cl) has -1.

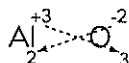
Therefore $+2 + (2 \times -1) = 0$. No charges appear in the overall formula because, once they are balanced, there is zero charge.

Using the cross method can help in writing formulas:

Place valency at the top of element



Cross over each valency number as subscripts



Remove valencies at top of element for final formula



3

	Cl^-	F^-	NO_3^-	O^{2-}	CO_3^{2-}	PO_4^{3-}
Na^+					Na_2CO_3	
H^+	HCl					
Li^+						Li_3PO_4
NH_4^+				$(\text{NH}_4)_2\text{O}$		
Mg^{2+}						
Ca^{2+}				CaO		
Al^{3+}	AlCl_3					

4

Formula	Name	Formula	Name
CaBr_2		FeCl_2	
HCl		HNO_3	
AlCl_3		Ag_2SO_4	
Na_3PO_4		NaHCO_3	
CaO		CuS	
$(\text{NH}_4)_2\text{CO}_3$		$\text{Fe}(\text{OH})_3$	

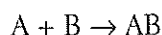


Name: _____

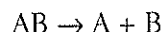
Skills: knowledge, interpreting

You should know the following reaction types. For more information about each type of reaction, see Focus 5.10.

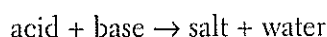
- **Combination:** Two or more elements combine to form one compound.



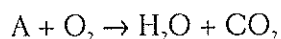
- **Decomposition:** The opposite of a combination reaction—a complex molecule breaks down to make simpler ones. These reactions have the general form:



- **Neutralisation:** An acid and a base react with each other. Generally, the product of this reaction is a salt and water:



- **Combustion:** Oxygen combines with a compound to form carbon dioxide and water. These reactions are exothermic, meaning they give out heat.



Questions

- Using the above information, classify the following reactions according to their type.

Equation	Reaction type
$2Fe(s) + O_2(g) \rightarrow 2FeO(s)$	
$H_2O(l) \rightarrow H_2(g) + O_2(g)$	
$Na_2CO_3(s) \rightarrow Na_2O(s) + CO_2(g)$	
$NaOH(aq) + HNO_3(aq) \rightarrow NaNO_3(aq) + H_2O(l)$	
$HCl(aq) + NaOH(aq) \rightarrow NaCl(aq) + H_2O(l)$	
$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g)$	
$Pb(s) + O_2(g) \rightarrow PbO_2(s)$	
$NH_4OH(aq) + HCl(aq) \rightarrow H_2O(l) + NH_4Cl(aq)$	
$Ca(OH)_2(aq) + HCl(aq) \rightarrow CaCl_2(aq) + H_2O(l)$	
$C_{10}H_8(l) + 12O_2(g) \rightarrow 10CO_2(g) + 4H_2O(g)$	

- Two of the above equations are not balanced. Identify the unbalanced equations, copy them into the space below and balance them.



Name: _____

3 For each of the following reactions:

- i Identify the type of reaction taking place
- ii Write a word equation for the reaction
- iii Write a balanced chemical equation for the reaction

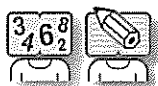
a Sulfuric acid reacts with potassium hydroxide to form water and a salt.

b Sulfur reacts with iron to form iron sulfide.

c Hydrogen peroxide (H_2O_2) breaks down to form oxygen gas and water.

d Barbecue gas (butane, C_4H_{10}) burns oxygen in an exothermic reaction.

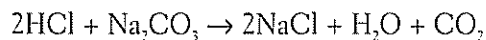
Studying a reaction



Name: _____

Skills: numeracy, interpreting

Hydrochloric acid reacts with sodium carbonate, producing sodium chloride, water and carbon dioxide gas. The balanced chemical equation for this reaction is:



A student conducted an experiment where 10 millilitres (mL) of acid (of constant concentration) was added to different masses of sodium carbonate. The student measured the volume of carbon dioxide gas produced in each case and recorded the results in a table:

Volume of HCl (mL)	Mass of Na_2CO_3 (g)	Volume of CO_2 produced (mL)
10.0	0.05	11.5
10.0	0.10	23.0
10.0	0.15	34.5
10.0	0.20	46.0
10.0	0.25	57.5
10.0	0.30	61.2
10.0	0.35	61.2
10.0	0.40	61.2
10.0	0.45	61.2

Questions

- 1 Plot a graph of the above data, showing the mass of sodium carbonate used on the horizontal axis, and the volume of carbon dioxide produced on the vertical axis.

- 2 Why do you think the volume of carbon dioxide produced did not increase after 0.3g of sodium carbonate was used?



Name: _____

Skills: knowledge, understanding**Part A**

Use the list of words below to fill in the blanks. Some words may be used more than once, and some not at all.

concentration	products	catalyst	smaller
increase	reactants	kinetic	rate
reactants	collisions	ratio	higher
surface area	used up		

- 1 The _____ of reaction is a measure of how quickly
_____ are turned into _____.

There are several ways to increase the rate of a reaction:

- 2 _____ the temperature. This will increase the
_____ energy of the reactants, causing more collisions. These collisions will
also have _____ energy and be more likely to cause a reaction.
- 3 Increase the _____ of a reactant. Having more reactant available causes
more successful _____ between particles.
- 4 Increase the _____ by breaking a
solid reactant into _____ pieces. This allows more collisions between the
reactants.
- 5 Add a _____ to speed up a reaction. A _____
helps a reaction proceed but is not _____
in the reaction.

Part B

Monica and Ethan designed an experiment to test the effect of temperature on the rate of reaction using hydrochloric acid and magnesium. They used the following method:

Equipment: test tubes, water bath, magnesium ribbon, thermometer, stop watch, ruler

Method

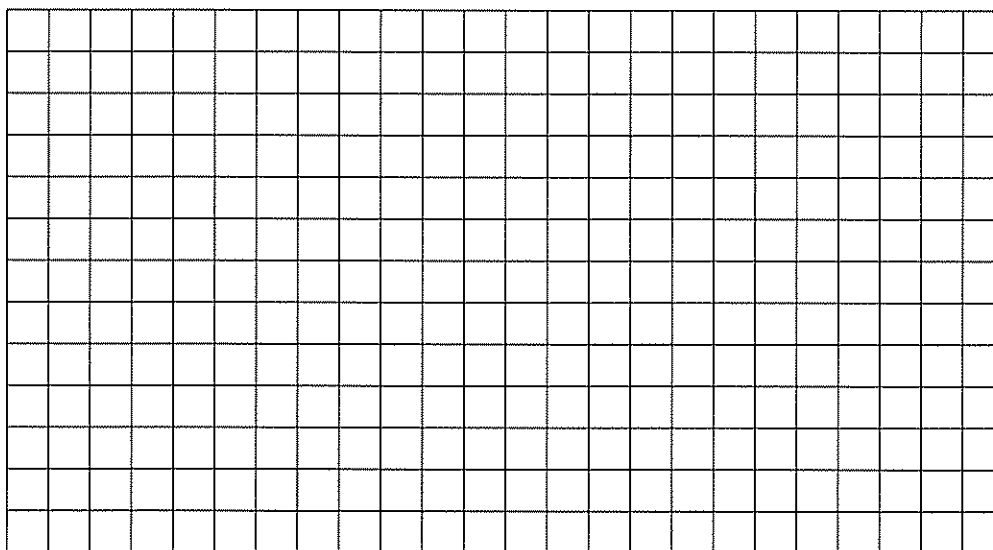
- Step 1 Add 20 ml of hydrochloric acid to a test tube.
- Step 2 Warm the test tube of acid to exactly 20°C using a water bath at 20°C.
- Step 3 When at the correct temperature add a 3 cm piece of magnesium ribbon to the test tube and start timing.
- Step 4 Stop timing when the magnesium is completely dissolved.
- Step 5 Repeat steps 1–4 at 40, 60, 80 and 100°C.

Name: _____

Results

Temperature (°C)	Time taken (seconds)
20	240
40	154
60	83
80	42
100	18

- 1 Plot these results on a graph, placing time on the horizontal (x) axis.



- 2 Describe any patterns and trends that you see in the results.

- 3 Use your graph to predict the time taken for magnesium to react at:

a 50°C _____

b 90°C _____

- 4 Use the results to draw a conclusion for the experiment.

- 5 Evaluate the experiment to decide whether it is a fair test.

- 6 Propose any improvements to the experimental method.

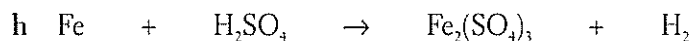
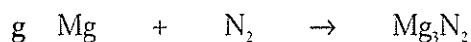
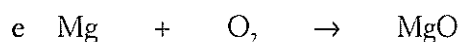
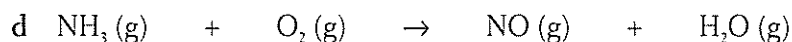
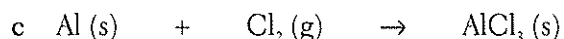
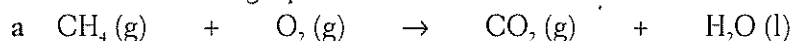


Writing and balancing chemical equations—page 1

Name: _____

Skills: interpreting, numeracy, knowledge

1 Balance the following equations:



2 Equations must be balanced so that the Law of Conservation of Mass holds true. State this law in your own words.

3 Use the information below to help answer the following questions.

Compound name	Compound formula	Compound name	Compound formula
Hydrochloric acid	HCl	Carbon dioxide	CO_2
Nitric acid	HNO_3	Calcium carbonate	CaCO_3
Sulfuric acid	H_2SO_4	Calcium nitrate	$\text{Ca}(\text{NO}_3)_2$
Magnesium chloride	MgCl_2	Magnesium hydroxide	$\text{Mg}(\text{OH})_2$
Barium sulfate	BaSO_4	Barium nitrate	$\text{Ba}(\text{NO}_3)_2$
Sodium sulfate	Na_2SO_4	Sodium hydroxide	NaOH
Water	H_2O	Sodium carbonate	Na_2CO_3

Write balanced chemical equations for each of the following reactions.

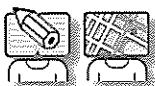
Hint: follow these steps:

- Write the word equation for the reaction.
- Directly underneath the word equation, write the unbalanced formula equation.
- Balance the equation.

Writing and balancing chemical equations — page 2

Name: _____

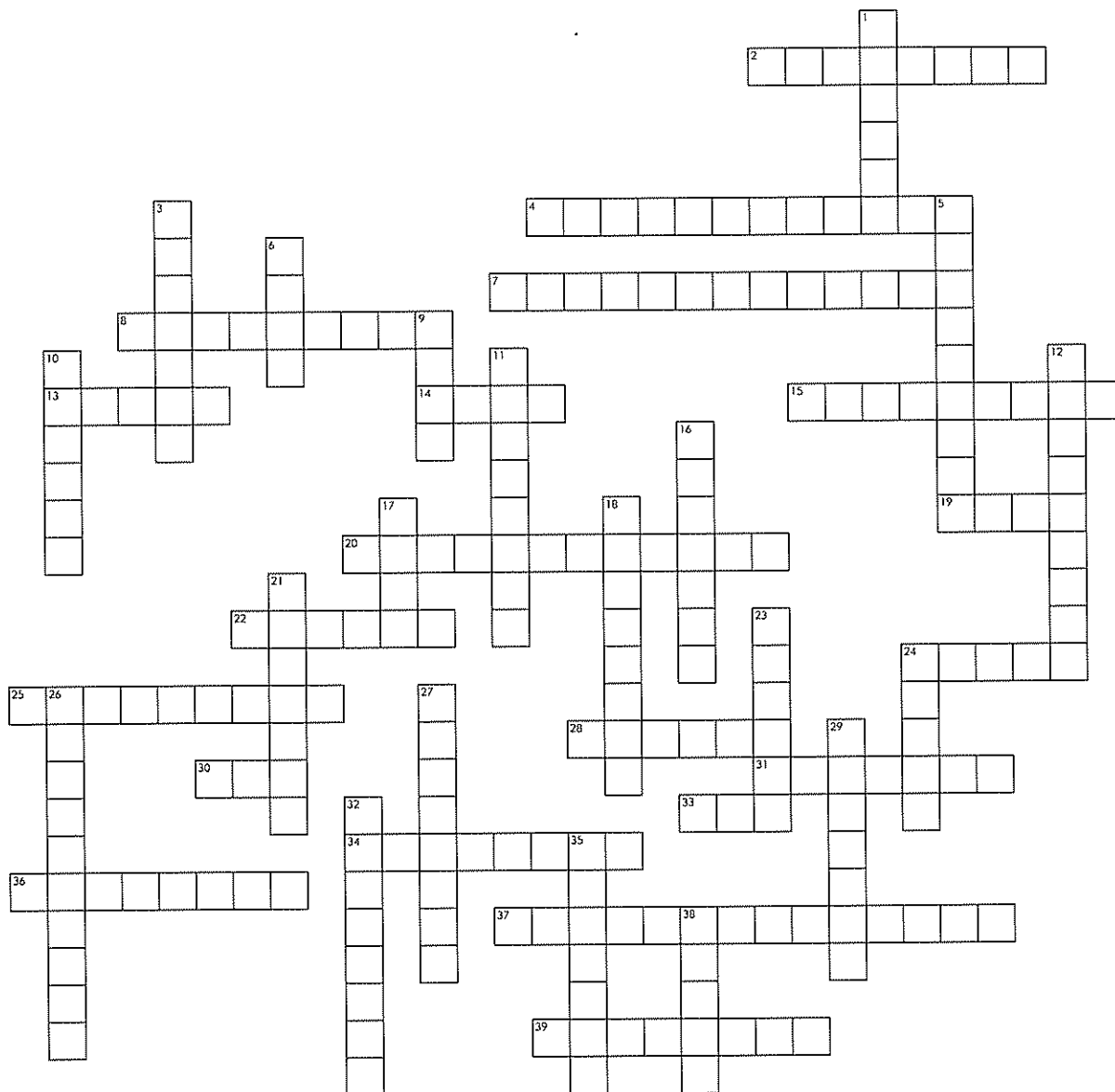
- a Hydrochloric acid is added to solid magnesium hydroxide, producing water and the soluble salt magnesium chloride.
- _____
- _____
- b Nitric acid is added to solid calcium carbonate, producing bubbles of carbon dioxide, water, and the soluble salt calcium nitrate.
- _____
- _____
- c Sodium hydroxide is added to sulfuric acid, producing water and the soluble salt sodium sulfate.
- _____
- _____
- d Sulfuric acid is poured over solid sodium carbonate, producing carbon dioxide, water and the soluble salt sodium sulfate.
- _____
- _____
- 4 Write balanced equations for the following reactions. This time you will need to write formulas first.
- a Iron metal reacts with chlorine gas to produce iron (III) chloride.
- _____
- b Sulfur dioxide gas reacts with oxygen to produce sulfur trioxide gas.
- _____



Natural and Processed Materials crossword—page 1

Name: _____

Skills: literacy, knowledge



Across

- 2 Colourless, odourless gas that is the lightest of all the chemical elements. It is also the most common element in the universe
- 4 Method of extraction that involves passing an electric current through a solution of the ore
- 7 Chemical opposite to composition reaction
- 8 Special substances that speed up the rate of a chemical reaction without themselves being used in the reaction
- 13 Type of bonding found in an ionic lattice
- 14 Smallest part that a chemical element can exist as
- 15 Can be hammered into thin sheets
- 19 Acid plus base produces a _____ plus water
- 20 If a large amount of acid is dissolved in water then the acid is said to be _____
- 22 Can be polished to a shiny finish
- 24 Alloy of copper and zinc used to make a variety of house fittings
- 25 These are found on the left-hand side of a chemical equation